

# Homework - 3

Consider the three colour channels of an RGB image as feature maps. Implement Spatial Pyramid Pooling (SPP) using max pooling to generate a fixed-length feature vector. Use pyramid levels consisting of  $1 \times 1$ ,  $2 \times 2$ ,  $4 \times 4$ , and  $8 \times 8$  spatial partitions (i.e., 1, 4, 16, and 64 regions). Demonstrate, through code implementation, how SPP generates feature vectors of identical length for two RGB images of different spatial dimensions.